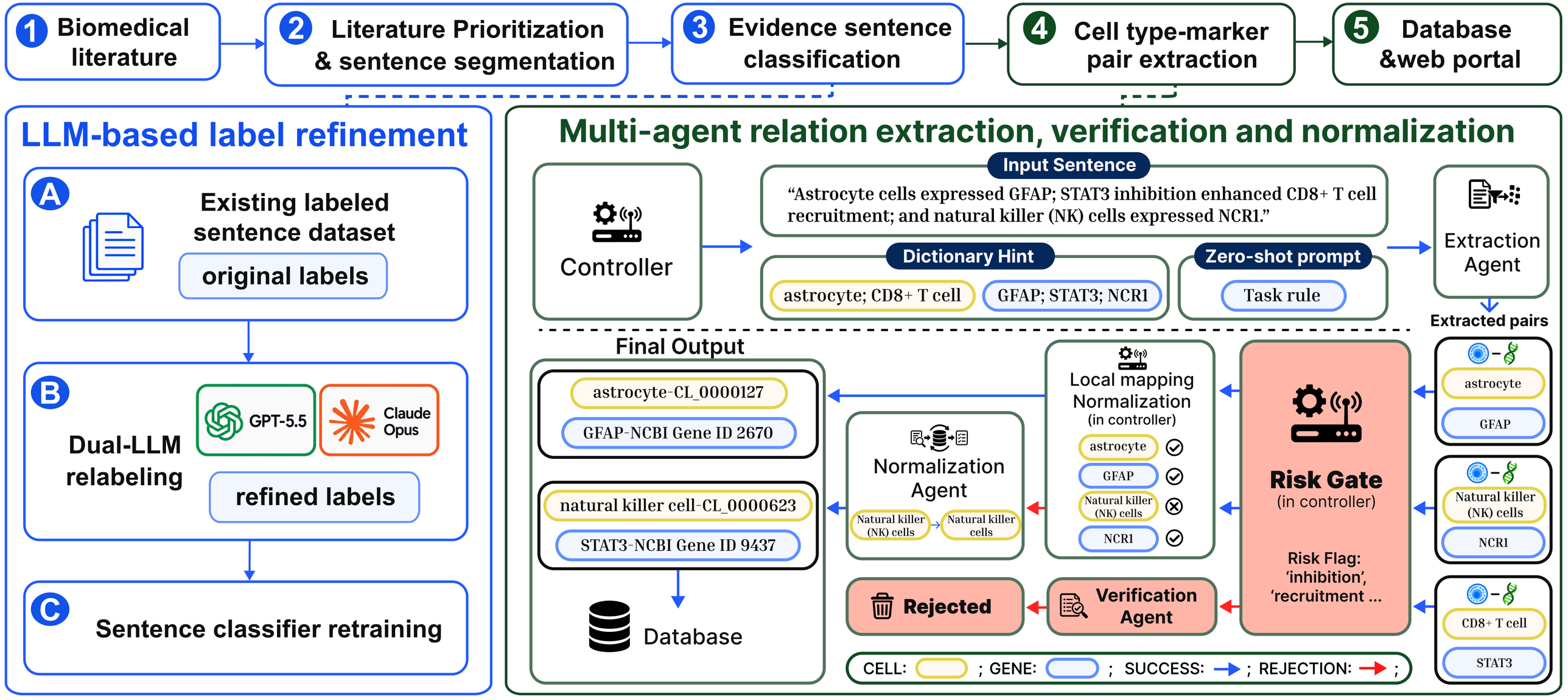

1 Biomedical literature
2 Literature Prioritization & sentence segmentation
3 Evidence sentence classification
4 Cell type-marker pair extraction
5 Database &web portal
LLM-based label refinement
A
Existing labeled sentence dataset
original labels
B
Dual-LLM relabeling
GPT-5.5
Claude Opus
refined labels
C
Sentence classifier retraining
Multi-agent relation extraction, verification and normalization
Controller
Input Sentence
“Astrocyte cells expressed GFAP; STAT3 inhibition enhanced CD8+ T cell recruitment; and natural killer (NK) cells expressed NCR1.”
Dictionary Hint
astrocyte; CD8+ T cell
GFAP; STAT3; NCR1
Zero-shot prompt
Task rule
Extraction Agent
Extracted pairs
astrocyte
GFAP
Natural killer (NK) cells
NCR1
CD8+ T cell
STAT3
Risk Gate
(in controller)
Risk Flag: ‘inhibition’, ‘recruitment ...
Local mapping Normalization
(in controller)
astrocyte
GFAP
Natural killer (NK) cells
NCR1
Normalization Agent
Natural killer (NK) cells
Natural killer cells
Verification Agent
Rejected
Final Output
astrocyte-CL_0000127
GFAP-NCBI Gene ID 2670
natural killer cell-CL_0000623
STAT3-NCBI Gene ID 9437
Database
CELL: ; GENE: ; SUCCESS: ; REJECTION: ;